Sigenergy

Modbus Protocol

Version: V2.9

Release date: 2026-05-13

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Contents

Version	Date	Change Description
V1.0-V1.8	2023-08-15 to 2024-08-05	Added description for interaction timeout. Supporting plant-wide power control. Added definition for alarm severity. Added a few phase power related registers and mode controlling registers. Added DC Charger related registers. Modified and added a few remote EMS and ESS control related registers. Added description of using different Modbus slave address querying different devices or power plant.
V2.0	2024-10-14	Added AC-Charger model and it's related registers. Added AC-Charger related system state and alarm appendix. Added DC-Charger related alarm appendix. Modified appendix names. Modified descriptions of RTU frame and PDU examples in chapter 6.
V2.1	2024-10-30	Added "Applicable model abbreviation" definition. Added applicable model columns in chapter 5. Added descriptions for communication interfaces. Added PV related registers. Modified alarm code names.
V2.2	2024-11-28	Added description for plant broadcast address. Added inverter level power control related registers. Modified plant parameter registers.
V2.3	2024-12-09	Added new applicable models.
V2.4	2025-02-05	Modified a few inverter's registers.

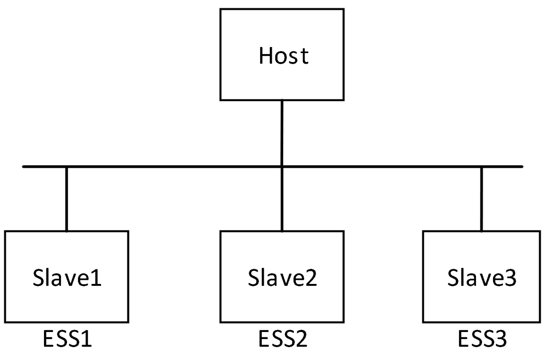
V2.5	2025-02-19	Added a few plant ESS related registers, two grid point and two PCS power control registers. Modified comments of a few holding registers. Added a few hybrid inverter battery temperature and voltage-related registers.
V2.6	2025-03-31	Added plant running info registers:plant PV total generation, Total load daily consumption, Total load consumption(30088-30094); Smart load 1-24 total consumption (30098-30144); smart load 1-24 power(30146-30192); Added plant parameter setting registers:[ESS] backup SOC, charge cut-off SOC and discharge cut-off SOC(40046 - 40048); Added hybrid inverter running info registers: ower adjustment related registers feedback value (30613-30619); PV daily generation and PV total generation (31509-31511).
V2.7	2025-05-23	Added plant running info registers: Third party inverter power(30194), Cumulative Energy Interface(30196-30268); Added an enumeration to register EMS working mode(30003): “5: Full Feed-in to Grid” and “9: Custom” Deleted hybrid inverter setting parameter: Grid code(40501) Appendix 1 “Running state” added an enumeration: Environmental Abnormality 0x07
V2.8	2025-11-28	Added plant running info registers: [Grid code]Rated Frequency(30276), [Grid code]Rated Voltage(30277), Current control command value(30279), Merged Alarm6(30280), Merged Alarm7(30281), General load power 30282, Total load power 30284. Added plant parameter setting registers: Active power regulation gradient(40049), Grid code Interface(40051-40068). Added hybrid inverter running info registers: pv voltage, pv current (31066-31105), [DC Charger] Running state 31513. Deleted hybrid inverter setting parameter: Remote EMS dispatch enable(41500) Added Appendix 12, Appendix 13 and Appendix 14. Added new applicable models.

V2.9	2026-05-13	Added plant running info registers: [ESS]Average cell temperature 30286. PV total daily generation 30272, PV total generation of previous day 30274 Added plant parameter setting registers: ESS Preheating tou Interface(50000-50183). PCC Power factor adjustment target value(Grid import) 40157, PCC Power factor adjustment target value(Grid export) 40158, Grid Power Loss Lockout Alarm Clear 40159 Added hybrid inverter running info registers: [DC Charger]dicharging registers 31514-31525. Added hybrid inverter setting parameter: [DC Charger]max charging power limit 41002, [DC Charger]max discharging power limit 41004 Added PSS and PID related registers. Added new applicable models. Added Appendix 15-21. Change "Address definition" to "Register definition" Added explanatory figures for the power limitation registers and power dispatch registers Added description of the series/types supported by DC chargers. Deleted Alarm severity level definition Deleted the note on "Applicable Model", The PV string counts shall be obtained from the register "String count 31025" in the inverter operation information table. Added enumeration value to register(40031)： 0x07 – Reserved, 0x08 – V2G (Vehicle to Grid) Replace all occurrences of "interface" with "register" in descriptions.

1. Introduction

This Modbus protocol complies the standard Modbus Application protocol specification. The physical media is multiple, such as RS485, Fast Ethernet, WLAN, Optical fiber and 4G. The figure below shows a simple host-slave mode in Modbus protocol.

Specifically, in a inverter-consisted power plant, to request information or control an individual devices, Modbus frames should be sent to the corresponding device's Modbus slave address, which must be set to a unique value among a inverter-consisted power plant in App. To request plant information or control plant behavior, Modbus frames should be sent to Modbus slave address 247, known as the "**plant address**". To control plant behavior and to not receive Modbus reply, Modbus frames should be sent to Modbus slave address 0, known as the "**plant broadcast address**".



2. Applicable Model

Table 2-1 lists the machine models applicable to this protocol. Each applicable model abbreviation in the table represents all the machine models listed to its right. In Chapter 5, "Register Address Definition," each register corresponds to a specific applicable model abbreviation, indicating that the register can only be read or written by the machine models the abbreviation represents.

Table 2-1 Applicable models

Applicable model abbreviation	Model
Hybrid Inv.	SigenStor EC (3.0, 3.6, 4.0, 4.6, 5.0, 6.0, 8.0, 10.0, 12.0) SP series
	Sigen Hybrid (3.0, 3.6, 4.0, 4.6, 5.0, 6.0) SP
	Sigen Hybrid (5.0, 6.0, 8.0, 10.0, 12.0, 15.0, 17.0, 20.0, 25.0, 30.0) TP series
	SigenStor EC (5.0, 6.0, 8.0, 10.0, 12.0, 15.0, 17.0, 20.0, 25.0, 30.0) TP/TPLV series
	Sigen PV (50, 60, 80, 99.9, 100, 110, 125)M1-HYA series
	PG Controller (3.8, 4.8, 5.7, 7.6, 9.6, 11.4) series
	Sigen PV (50, 60, 80, 99.9, 100, 110,125)M1-HYB series
	Sigen Hybrid (2.0, 2.5, 3.0, 3.6, 4.0, 4.6, 5.0, 6.0) SP2series
	Sigen Hybrid (3.0, 4.0, 5.0, 6.0, 8.0, 10.0, 12.0) TP2series
EVAC	Sigen EVAC (7, 11, 22) 4G T2 WH
	Sigen EVAC (7, 11, 22) 4G T2SH WH
	PG EVAC (9.6, 11.5) series
PV Inv.	Sigen PV Max (3.0, 3.6, 4.0, 4.6, 5.0, 6.0) SP
	Sigen PV Max (5.0, 6.0, 8.0, 10.0, 12.0, 15.0, 17.0, 20.0, 25.0) TP
	Sigen PV (50, 60, 80, 99.9, 100, 110, 125)M1 series
	Sigen PV (500)H1 series
	Sigen PV (370, 400)H2 series
Logger	Sigen Logger A1 series
Commbox	Sigen Commbox (A05, A06, A07, A08, A09) series

3. Communication Interface

3.1 RS485

For applicable model abbreviation type "EVAC" devices, the interface described in this section is not supported.

For applicable model abbreviation type "Hybrid Inv." and “PV Inv.” devices, third-party controllers only need to connect to one device in the power plant through the interface (RS485) described in this section and can read or write registers of all devices within the power plant (with different modbus slave addresses). The power plant can consist of multiple parallel-connected "Hybrid Inv." or “PV Inv.” devices.

Table 3-1 RS485 interface description

Parameter	Description
Transfer mode	RTU mode
Communication mode	Half duplex
Baud rate	9600bps(default)
Start bit	1
Data bit	8
Check bit	None
Stop bit	1

3.2 Fast Ethernet/WLAN/Optical fiber/4G

For applicable model abbreviation type "Hybrid Inv." and “PV Inv.” devices, third-party controllers only need to connect to one device in the power plant through the interface described in this section and can read or write registers of all devices within the power plant (with different modbus slave addresses). The power plant can consist of multiple parallel-connected "Hybrid Inv." or "PV Inv." devices.

For applicable model abbreviation type "EVAC" device, it must be connected to a "Hybrid Inv." or "PV Inv."device. Third-party controllers have to connect to the "Hybrid Inv." or "PV Inv." device using the interface described in this section to access registers of the "EVAC" device.

Table 3-2 Fast Ethernet/WLAN/Optical fiber/4G interface description

Parameter	Description
Transfer mode	TCP mode
Communication mode	Full duplex
Link layer Mode	TCP Server
Application layer Mode	Slave
Port	502

3.3 Fast Ethernet/WLAN/Optical fiber/4G*

For applicable model abbreviation type "Hybrid Inv." and "PV Inv." devices, third-party controllers only need to connect to one device in the power plant through the interface described in this section and can read or write registers of all devices within the power plant (with different modbus slave addresses). The power plant can consist of multiple parallel-connected "Hybrid Inv." or "PV

Inv." devices.

For applicable model abbreviation type "EVAC" device, it must be connected to a "Hybrid Inv." or "PV Inv." device. Third-party controllers have to connect to the "Hybrid Inv." or "PV Inv." device using the interface described in this section to access registers of the "EVAC" device.

Table 3-3 Fast Ethernet/WLAN/Optical fiber/4G interface description

Parameter	Description
Transfer mode	TCP mode
Communication mode	Full duplex
Link layer Mode	TCP Client
Application layer Mode	Slave
Port	custom

*Note :To be specific, if 4G is the only physical communication media, the protocol then only supports one inverter device to connect the third party cloud as a client.

4. Technical Terms

4.1 Technical item name specification:

Table 4-1 Technical item description

Item	Description
Host	The one that initiates an application request is referred to the host
Slave	The one that responds to an application request is referred to the slave
Access plant address	247
Plant broadcast address	0

Slave address range	1-246
U16	Unsigned integer of 16-bit
U32	Unsigned integer of 32-bit
U64	Unsigned integer of 64-bit
S16	Signed integer of 16-bit
S32	Signed integer of 32-bit
STRING	Character string in ASCII
BIT16	Bit-mapped integer
RO	Read only, only support 0x04 command
WO	Write only, only support 0x06 command
RW	Read and write, support 0x04、0x06、0x10 command

4.2 Interaction timeout

A communication process following the Modbus protocol should always be stared by a host.

Minimum Request period : 1000 ms

After sending an unicast request, before receiving a respond from the slave device, the host should wait for up to 1000ms to send a new unicast request to the slave device. If no respond is received from the slave device after waiting for 1000 ms, the host should regard this request as a timeout. In poor network conditions or when using extra-long RS485 connections, it may be necessary to appropriately increase the minimum request period.

Plant broadcast address:

When the host sends a broadcast request to Modbus slave address 0, the devices will perform but will not reply Modbus frame.

5. Register Definition

5.1 Plant running information register definition (input register)

The registers below can only be accessed by slave address 247, namely “plant address”. To obtain power plant data, inquiries should be send to address 247.

Table 5-1 Plant running information register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Hybrid Inv,	PV inv.	Comment
1	System time	30000	2	RO	U32	1	s	√	√	Epoch seconds
2	System time zone	30002	1	RO	S16	1	min	√	√	
3	EMS work mode	30003	1	RO	U16	N/A	N/A	√	√	0: Max self consumption; 1: AI Mode; 2: TOU 5: Full Feed-in to Grid 7: Remote EMS mode 9: Custom
4	[Grid Sensor] Status	30004	1	RO	U16	N/A	N/A	√	√	(gateway or meter connection status) 0: not connected 1: connected
5	[Grid sensor] Active power	30005	2	RO	S32	1000	kW	√	√	Data collected from grid sensor at grid to system checkpoint; >0 buy from grid; <0 sell to grid
6	[Grid sensor] Reactive power	30007	2	RO	S32	1000	kVar	√	√	Data collected from grid sensor at grid to system checkpoint;

7	On/Off Grid status	30009	1	RO	U16	N/A	N/A	√		0: on grid 1: off grid (auto) 2: off grid (manual)
8	Max active power	30010	2	RO	U32	1000	kW	√	√	This is should be the base value of all active power adjustment actions
9	Max apparent power	30012	2	RO	U32	1000	kVar	√	√	This is should be the base value of all reactive power adjustment actions
10	[ESS] SOC	30014	1	RO	U16	10	%	√		
11	Plant phase A active power	30015	2	RO	S32	1000	kW	√	√	
12	Plant phase B active power	30017	2	RO	S32	1000	kW	√	√	
13	Plant phase C active power	30019	2	RO	S32	1000	kW	√	√	
14	Plant phase A reactive power	30021	2	RO	S32	1000	kVar	√	√	
15	Plant phase B reactive power	30023	2	RO	S32	1000	kVar	√	√	
16	Plant phase C reactive power	30025	2	RO	S32	1000	kVar	√	√	
17	Merged Alarm1	30027	1	RO	U16	N/A	N/A	√	√	If any hybrid inverter has alarm , then this alarm will be set accordingly. Refer to Appendix 2
18	Merged Alarm2	30028	1	RO	U16	N/A	N/A	√	√	If any hybrid inverter has alarm , then this alarm will be set accordingly. Refer to Appendix 3
19	Merged Alarm3	30029	1	RO	U16	N/A	N/A	√		If any hybrid inverter has alarm , then this alarm will be set accordingly. Refer to Appendix 4
20	Merged Alarm4	30030	1	RO	U16	N/A	N/A	√	√	If any hybrid inverter has alarm , then this alarm will be set accordingly. Refer to Appendix 5
21	Plant active power	30031	2	RO	S32	1000	kW	√	√	
22	Plant reactive power	30033	2	RO	S32	1000	kVar	√	√	

23	Photovoltaic power	30035	2	RO	S32	1000	kW	√	√	
24	[ESS] Power	30037	2	RO	S32	1000	kW	√		<0: discharging >0: charging
25	Available max active power	30039	2	RO	U32	1000	kW	√	√	Feed to the ac terminal. Count only the running inverters
26	Available min active power	30041	2	RO	U32	1000	kW	√		Absorb from the ac terminal. Count only the running inverters
27	Available max reactive power	30043	2	RO	U32	1000	kVar	√	√	Feed to the ac terminal. Count only the running inverters
28	Available min reactive power	30045	2	RO	U32	1000	kVar	√	√	Absorb from the ac terminal. Count only the running inverters
29	[ESS]Available max charging power	30047	2	RO	U32	1000	kW	√		Count only the running inverters
30	[ESS]Available max discharging power	30049	2	RO	U32	1000	kW	√		Count only the running inverters
31	Plant running state	30051	1	RO	U16	N/A	N/A	√	√	Refer to Appendix 1
32	[Grid sensor] Phase A active power	30052	2	RO	S32	1000	kW	√	√	Data collected from grid sensor at grid to system checkpoint; >0 buy from grid; <0 sell to grid
33	[Grid sensor] Phase B active power	30054	2	RO	S32	1000	kW	√	√	Data collected from grid sensor at grid to system checkpoint; >0 buy from grid; <0 sell to grid
34	[Grid sensor] Phase C active power	30056	2	RO	S32	1000	kW	√	√	Data collected from grid sensor at grid to system checkpoint; >0 buy from grid; <0 sell to grid
35	[Grid sensor] Phase A reactive power	30058	2	RO	S32	1000	kVar	√	√	Data collected from grid sensor at grid to system checkpoint;

36	[Grid sensor] Phase B reactive power	30060	2	RO	S32	1000	kVar	√	√	Data collected from grid sensor at grid to system checkpoint;
37	[Grid sensor] Phase C reactive power	30062	2	RO	S32	1000	kVar	√	√	Data collected from grid sensor at grid to system checkpoint;
38	[ESS]Available max charging capacity	30064	2	RO	U32	100	kWh	√		Count only the running inverters
39	[ESS]Available max discharging capacity	30066	2	RO	U32	100	kWh	√		Count only the running inverters
40	[ESS] Rated charging power	30068	2	RO	U32	1000	kW	√		
41	[ESS] Rated discharging power	30070	2	RO	U32	1000	kW	√		
42	Merged Alarm5	30072	1	RO	U16	N/A	N/A	√		If any hybrid inverter has alarm , then this alarm will be set accordingly. Refer to Appendix 11
43	Reserved	30073	10	RO	N/A	N/A	N/A			
44	[ESS] Rated energy capacity	30083	2	RO	U32	100	kWh	√		
45	[ESS] Charge Cut-Off SOC	30085	1	RO	U16	10	%	√		
46	[ESS] Discharge Cut-Off SOC	30086	1	RO	U16	10	%	√		
47	[ESS] SOH	30087	1	RO	U16	10	%	√		This value is the weighted average of the SOH of all ESS devices in the power plant, with each rated capacity as the weight.
48	Plant PV total generation	30088	4	RO	U64	100	kWh	√	√	
49	Total load daily consumption	30092	2	RO	U32	100	kWh	√	√	
50	Total load consumption	30094	4	RO	U64	100	kWh	√	√	
51	[Smart load 1] Total consumption	30098	2	RO	U32	100	kWh	√	√	
52	[Smart load 2] Total consumption	30100	2	RO	U32	100	kWh	√	√	

53	[Smart load 3] Total consumption	30102	2	RO	U32	100	kWh	√	√	
54	[Smart load 4] Total consumption	30104	2	RO	U32	100	kWh	√	√	
55	[Smart load 5] Total consumption	30106	2	RO	U32	100	kWh	√	√	
56	[Smart load 6] Total consumption	30108	2	RO	U32	100	kWh	√	√	
57	[Smart load 7] Total consumption	30110	2	RO	U32	100	kWh	√	√	
58	[Smart load 8] Total consumption	30112	2	RO	U32	100	kWh	√	√	
59	[Smart load 9] Total consumption	30114	2	RO	U32	100	kWh	√	√	
60	[Smart load 10] Total consumption	30116	2	RO	U32	100	kWh	√	√	
61	[Smart load 11] Total consumption	30118	2	RO	U32	100	kWh	√	√	
62	[Smart load 12] Total consumption	30120	2	RO	U32	100	kWh	√	√	
63	[Smart load 13] Total consumption	30122	2	RO	U32	100	kWh	√	√	
64	[Smart load 14] Total consumption	30124	2	RO	U32	100	kWh	√	√	
65	[Smart load 15] Total consumption	30126	2	RO	U32	100	kWh	√	√	
66	[Smart load 16] Total consumption	30128	2	RO	U32	100	kWh	√	√	
67	[Smart load 17] Total consumption	30130	2	RO	U32	100	kWh	√	√	
68	[Smart load 18] Total consumption	30132	2	RO	U32	100	kWh	√	√	
69	[Smart load 19] Total consumption	30134	2	RO	U32	100	kWh	√	√	
70	[Smart load 20] Total consumption	30136	2	RO	U32	100	kWh	√	√	
71	[Smart load 21] Total consumption	30138	2	RO	U32	100	kWh	√	√	
72	[Smart load 22] Total consumption	30140	2	RO	U32	100	kWh	√	√	
73	[Smart load 23] Total consumption	30142	2	RO	U32	100	kWh	√	√	

74	[Smart load 24] Total consumption	30144	2	RO	U32	100	kWh	√	√	
75	[Smart load 1] Power	30146	2	RO	S32	1000	kW	√	√	
76	[Smart load 2] Power	30148	2	RO	S32	1000	kW	√	√	
77	[Smart load 3] Power	30150	2	RO	S32	1000	kW	√	√	
78	[Smart load 4] Power	30152	2	RO	S32	1000	kW	√	√	
79	[Smart load 5] Power	30154	2	RO	S32	1000	kW	√	√	
80	[Smart load 6] Power	30156	2	RO	S32	1000	kW	√	√	
81	[Smart load 7] Power	30158	2	RO	S32	1000	kW	√	√	
82	[Smart load 8] Power	30160	2	RO	S32	1000	kW	√	√	
83	[Smart load 9] Power	30162	2	RO	S32	1000	kW	√	√	
84	[Smart load 10] Power	30164	2	RO	S32	1000	kW	√	√	
85	[Smart load 11] Power	30166	2	RO	S32	1000	kW	√	√	
86	[Smart load 12] Power	30168	2	RO	S32	1000	kW	√	√	
87	[Smart load 13] Power	30170	2	RO	S32	1000	kW	√	√	
88	[Smart load 14] Power	30172	2	RO	S32	1000	kW	√	√	
89	[Smart load 15] Power	30174	2	RO	S32	1000	kW	√	√	
90	[Smart load 16] Power	30176	2	RO	S32	1000	kW	√	√	
91	[Smart load 17] Power	30178	2	RO	S32	1000	kW	√	√	
92	[Smart load 18] Power	30180	2	RO	S32	1000	kW	√	√	
93	[Smart load 19] Power	30182	2	RO	S32	1000	kW	√	√	
94	[Smart load 20] Power	30184	2	RO	S32	1000	kW	√	√	

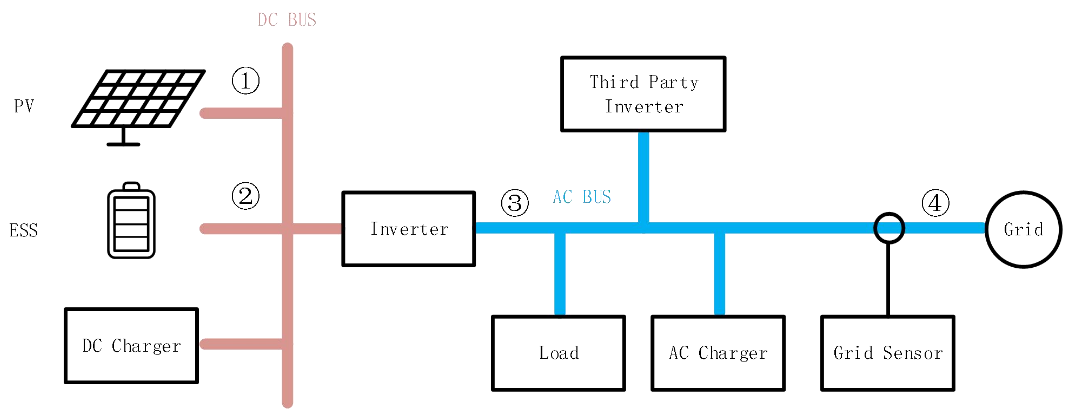
95	[Smart load 21] Power	30186	2	RO	S32	1000	kW	√	√	
96	[Smart load 22] Power	30188	2	RO	S32	1000	kW	√	√	
97	[Smart load 23] Power	30190	2	RO	S32	1000	kW	√	√	
98	[Smart load 24] Power	30192	2	RO	S32	1000	kW	√	√	
99	Third Party inverter active power	30194	2	RO	S32	1000	kW	√	√	
100	Total generation of third party inverter	30196	4	RO	U64	100	kWh	√	√	
101	Total charged energy of the ESS	30200	4	RO	U64	100	kWh	√	√	
102	Total discharged energy of the ESS	30204	4	RO	U64	100	kWh	√	√	
103	Total charged energy of the EVDC	30208	4	RO	U64	100	kWh	√	√	
104	Total discharged energy of the EVDC	30212	4	RO	U64	100	kWh	√	√	
105	Total imported energy	30216	4	RO	U64	100	kWh	√	√	
106	Total exported energy	30220	4	RO	U64	100	kWh	√	√	
107	Total energy output of oil-fueled generator	30224	4	RO	U64	100	kWh	√	√	
108	Total energy consumption of common loads	30228	4	RO	U64	100	kWh	√	√	New statistics interface, excluding EVAC and EVDC charged energy.
109	Total charged energy of the EVAC	30232	4	RO	U64	100	kWh	√	√	New statistics interface,
110	Total generation of self PV	30236	4	RO	U64	100	kWh	√	√	New statistics interface, excluding generation of third party inverter
111	Total generation of third party inverter	30240	4	RO	U64	100	kWh	√	√	New statistics interface
112	Total charged energy of the ESS	30244	4	RO	U64	100	kWh	√	√	New statistics interface

113	Total discharged energy of the ESS	30248	4	RO	U64	100	kWh	√	√	New statistics interface
114	Total charged energy of the EVDC	30252	4	RO	U64	100	kWh	√	√	New statistics interface
115	Total discharged energy of the EVDC	30256	4	RO	U64	100	kWh	√	√	New statistics interface
116	Total imported energy	30260	4	RO	U64	100	kWh	√	√	New statistics interface
117	Total exported energy	30264	4	RO	U64	100	kWh	√	√	New statistics interface
118	Total energy output of oil-fueled generator	30268	4	RO	U64	100	kWh	√	√	New statistics interface
119	PV total daily generation	30272	2	RO	U32	100	kWh	√	√	
120	PV total generation of previous day	30274	2	RO	U32	100	kWh	√	√	
121	[Grid code]Rated Frequency	30276	1	RO	U16	100	Hz	√	√	
122	[Grid code]Rated Voltage	30277	2	RO	U32	100	V	√	√	
123	Current control command value	30279	1	RO	U16	100	%	√	√	Use of Remote Output Control in Japan
124	Merged Alarm6	30280	1	RO	U16	NA	NA	√	√	Refer to Appendix 12
125	Merged Alarm7	30281	1	RO	U16	NA	NA	√	√	Refer to Appendix 13
126	General load power	30282	2	RO	S32	1000	kW	√	√	
127	Total load power	30284	2	RO	S32	1000	kW	√	√	
128	[ESS]Average cell temperature	30286	1	RO	S16	10	℃	√		

*Note :For the new energy-statistics interface (registers 30228~30268), after upgrading the device firmware to support this interface, the register values will reset to 0 and start fresh counting without inheriting historical data.

5.2 Plant parameter setting register definition (holding register)

The registers below can only be accessed by slave address 0 or 247. To modify plant-level registers, send commands to address 0 or 247. When sending commands to address 0, the device will only execute and will not reply. When sending commands to address 247, the device will both execute and respond.



*Note1 : ESS only SigenStor, Sigen Hybrid, Sigen PV M1-HYB series support.
*Note2 : DC Charger only SigenStor, Sigen Hybrid support.
*Note3: The Modbus registers related to the above components are only valid in the supported series.

5.2.1 Plant parameter setting power limitation register definition (holding register)

Table 5-2-1 Plant setting parameter power limitation register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Hybrid Inv,	PV inv.	Comment
1	ESS max charging limit	40032	2	RW	U32	1000	kW	√		[0, Rated ESS charging power]. Takes effect globally regardless of the EMS operating mode.
2	ESS max discharging limit	40034	2	RW	U32	1000	kW	√		[0, Rated ESS discharging power].

										Takes effect globally regardless of the EMS operating mode.
3	PV max power limit	40036	2	RW	U32	1000	kW	√		Takes effect globally regardless of the EMS operating mode.
4	[Grid Point]Maximum export limitation	40038	2	RW	U32	1000	kW	√	√	Grid Sensor needed. Takes effect globally regardless of the EMS operating mode.
5	[Grid Point] Maximum import limitation	40040	2	RW	U32	1000	kW	√	√	Grid Sensor needed. Takes effect globally regardless of the EMS operating mode.
6	PCS maximum export limitation	40042	2	RW	U32	1000	kW	√	√	Range:[0, 0xFFFFFFFE]。With value 0xFFFFFFFF, register is not valid. In all other cases, Takes effect globally.
7	PCS maximum import limitation	40044	2	RW	U32	1000	kW	√	√	Range:[0, 0xFFFFFFFE]。With value 0xFFFFFFFF, register is not valid. In all other cases, Takes effect globally.
8	PCC Power factor adjustment target value (Grid import)	40157	1	RW	S16	1000	N/A	√	√	Range: (-1, -0.8] U [0.8, 1]. Grid Sensor needed.Takes effect upon grid import.
9	PCC Power factor adjustment target value (Grid export)	40158	1	RW	S16	1000	N/A	√	√	Range: (-1, -0.8] U [0.8, 1].Grid Sensor needed.Takes effect upon grid export.

*Note1 : Without grid sensor, The control objective of the registers 40157 and 40158 has been changed from "PCC Power" to "PCS Power".

*Note2 : Register 40157 and 40158 only Sigen PV M1-HYB series support.

As shown in the above figure, the specific points where the power limitation registers take effect are as follows:

(1) PV max power limit, The command applies at the location marked ① in figure, meaning that the maximum output power of the PV strings in the plant must not exceed the value configured in this register.

(2) ESS max charging limit, The command applies at the location marked ② in figure, meaning that the maximum charging power of the ESS in the plant must not exceed the value configured in this register.

(3) ESS max discharging limit, The command applies at the location marked ② in figure, meaning that the maximum discharging power of the ESS in the plant must not exceed the value configured in this register.

(4) PCS maximum export limitation, The command applies at the location marked ③ in figure, meaning that the maximum active power output of the inverter in the plant must not exceed the value configured in this register.

(5) PCS maximum import limitation, The command applies at the location marked ③ in figure, meaning that the maximum active power input of the inverter in the plant must not exceed the value configured in this register.

(6) [Grid Point]Maximum export limitation, The command applies at the location marked ④ in figure, meaning that the maximum active power fed into the grid must not exceed the value configured in this register.

(7) [Grid Point]Maximum import limitation, The command applies at the location marked ④ in figure, meaning that the maximum active power absorbed from the grid must not exceed the value configured in this register.

(8) PCC Power factor adjustment target value (Grid import), The command applies at the location marked ④ in figure, meaning that when purchasing electricity at the grid connection point, the power factor must be controlled to the value configured in this register.

(9) PCC Power factor adjustment target value (Grid export), The command applies at the location marked ④ in figure, meaning that when selling electricity at the grid connection point, the power factor must be controlled to the value configured in this register.

5.2.2 Plant parameter setting power dispatch register definition (holding register)

The power dispatch registers in Table 5-2-2 applies at the location marked ③ in figure. For these registers to take effect, the following two conditions must be met:
a. The Modbus register for remote EMS enable (40029) is set to 0x01;
b. The Modbus register for remote EMS control mode (40031) is set to 0x00.
*Note : These registers only control the power of PCS and is independent of loads and grid sensors.

Table 5-2-2 Plant setting parameter power dispatch register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Hybrid Inv,	PV inv.	Comment
1	Active power fixed adjustment target value	40001	2	RW	S32	1000	kW	√	√	
2	Reactive power fixed adjustment target value	40003	2	RW	S32	1000	kVar	√	√	Range: [-60.00 * base value ,60.00 * base value].
3	Active power percentage adjustment target value	40005	1	RW	S16	100	%	√	√	Range: [-100.00,100.00]
4	Q/S adjustment target value	40006	1	RW	S16	100	%	√	√	Range: [-60.00,60.00].
5	Power factor adjustment target value	40007	1	RW	S16	1000	N/A	√	√	Range: (-1, -0.8] U [0.8, 1].
6	Phase A active power fixed adjustment target value	40008	2	RW	S32	1000	kW	√	√	Valid only when output type is L1/L2/L3/N
7	Phase B active power fixed adjustment target value	40010	2	RW	S32	1000	kW	√	√	Valid only when output type is L1/L2/L3/N

8	Phase C active power fixed adjustment target value	40012	2	RW	S32	1000	kW	√	√	Valid only when output type is L1/L2/L3/N
9	Phase A reactive power fixed adjustment target value	40014	2	RW	S32	1000	kVar	√	√	Valid only when output type is L1/L2/L3/N
10	Phase B reactive power fixed adjustment target value	40016	2	RW	S32	1000	kVar	√	√	Valid only when output type is L1/L2/L3/N
11	Phase C reactive power fixed adjustment target value	40018	2	RW	S32	1000	kVar	√	√	Valid only when output type is L1/L2/L3/N
12	Phase A Active power percentage adjustment target value	40020	1	RW	S16	100	%	√	√	Valid only when output type is L1/L2/L3/N. Range: [-100.00,100.00]
13	Phase B Active power percentage adjustment target value	40021	1	RW	S16	100	%	√	√	Valid only when output type is L1/L2/L3/N. Range: [-100.00,100.00]
14	Phase C Active power percentage adjustment target value	40022	1	RW	S16	100	%	√	√	Valid only when output type is L1/L2/L3/N. Range: [-100.00,100.00]
15	Phase A Q/S fixed adjustment target value	40023	1	RW	S16	100	%	√	√	Valid only when output type is L1/L2/L3/N. Range: [-60.00,60.00]
16	Phase B Q/S fixed adjustment target value	40024	1	RW	S16	100	%	√	√	Valid only when output type is L1/L2/L3/N. Range: [-60.00,60.00]
17	Phase C Q/S fixed adjustment target value	40025	1	RW	S16	100	%	√	√	Valid only when output type is L1/L2/L3/N. Range: [-60.00,60.00]

5.2.3 Plant parameter setting ESS Preheating register definition (holding register)

Table 5-2-3 Plant setting parameter ESS Preheating register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Hybrid Inv,	PV inv.	Comment
1	[ESS Preheating] Preheating Enable	50000	1	RW	U16	N/A	N/A	√		0: Disable 1: Enable

2	[ESS Preheating] Preheating mode	50001	1	RW	U16	N/A	N/A	√		0: Automatic 1: Manual
3	[ESS Preheating] Advance Preheating Enable	50002	1	RW	U16	N/A	N/A	√		0: Disable 1: Enable Takes effect when Preheating mode is manual.
4	[ESS Preheating] tou 1:Start Time	50003	2	RW	U32	1	s	√		Epoch seconds with time zone to local time
5	[ESS Preheating] tou 1:End Time	50005	2	RW	U32	1	s	√		Epoch seconds with time zone to local time
6	[ESS Preheating] tou 1:Target Charging/ Discharging Power	50007	2	RW	S32	1000	kW	√		<0: discharging >0: charging
7	[ESS Preheating] tou 2:Start Time	50009	2	RW	U32	1	s	√		
8	[ESS Preheating] tou 2:End Time	50011	2	RW	U32	1	s	√		
9	[ESS Preheating] tou 2:Target Charging/ Discharging Power	50013	2	RW	S32	1000	kW	√		
10	[ESS Preheating] tou 3:Start Time	50015	2	RW	U32	1	s	√		
11	[ESS Preheating] tou 3:End Time	50017	2	RW	U32	1	s	√		
12	[ESS Preheating] tou 3:Target Charging/ Discharging Power	50019	2	RW	S32	1000	kW	√		
13	[ESS Preheating] tou 4:Start Time	50021	2	RW	U32	1	s	√		
14	[ESS Preheating] tou 4:End Time	50023	2	RW	U32	1	s	√		
15	[ESS Preheating] tou 4:Target Charging/ Discharging Power	50025	2	RW	S32	1000	kW	√		
16	[ESS Preheating] tou 5:Start Time	50027	2	RW	U32	1	s	√		
17	[ESS Preheating] tou 5:End Time	50029	2	RW	U32	1	s	√		
18	[ESS Preheating] tou 5:Target Charging/ Discharging Power	50031	2	RW	S32	1000	kW	√		

19	[ESS Preheating] tou 6:Start Time	50033	2	RW	U32	1	s	√		
20	[ESS Preheating] tou 6:End Time	50035	2	RW	U32	1	s	√		
21	[ESS Preheating] tou 6:Target Charging/ Discharging Power	50037	2	RW	S32	1000	kW	√		
22	[ESS Preheating] tou 7:Start Time	50039	2	RW	U32	1	s	√		
23	[ESS Preheating] tou 7:End Time	50041	2	RW	U32	1	s	√		
24	[ESS Preheating] tou 7:Target Charging/ Discharging Power	50043	2	RW	S32	1000	kW	√		
25	[ESS Preheating] tou 8:Start Time	50045	2	RW	U32	1	s	√		
26	[ESS Preheating] tou 8:End Time	50047	2	RW	U32	1	s	√		
27	[ESS Preheating] tou 8:Target Charging/ Discharging Power	50049	2	RW	S32	1000	kW	√		
28	[ESS Preheating] tou 9:Start Time	50051	2	RW	U32	1	s	√		
29	[ESS Preheating] tou 9:End Time	50053	2	RW	U32	1	s	√		
30	[ESS Preheating] tou 9:Target Charging/ Discharging Power	50055	2	RW	S32	1000	kW	√		
31	[ESS Preheating] tou 10:Start Time	50057	2	RW	U32	1	s	√		
32	[ESS Preheating] tou 10:End Time	50059	2	RW	U32	1	s	√		
33	[ESS Preheating] tou 10:Target Charging/ Discharging Power	50061	2	RW	S32	1000	kW	√		
34	[ESS Preheating] tou 11:Start Time	50063	2	RW	U32	1	s	√		
35	[ESS Preheating] tou 11:End Time	50065	2	RW	U32	1	s	√		
36	[ESS Preheating] tou 11:Target Charging/ Discharging Power	50067	2	RW	S32	1000	kW	√		

37	[ESS Preheating] tou 12:Start Time	50069	2	RW	U32	1	s	√		
38	[ESS Preheating] tou 12:End Time	50071	2	RW	U32	1	s	√		
39	[ESS Preheating] tou 12:Target Charging/ Discharging Power	50073	2	RW	S32	1000	kW	√		
40	[ESS Preheating] tou 13:Start Time	50075	2	RW	U32	1	s	√		
41	[ESS Preheating] tou 13:End Time	50077	2	RW	U32	1	s	√		
42	[ESS Preheating] tou 13:Target Charging/ Discharging Power	50079	2	RW	S32	1000	kW	√		
43	[ESS Preheating] tou 14:Start Time	50081	2	RW	U32	1	s	√		
44	[ESS Preheating] tou 14:End Time	50083	2	RW	U32	1	s	√		
45	[ESS Preheating] tou 14:Target Charging/ Discharging Power	50085	2	RW	S32	1000	kW	√		
46	[ESS Preheating] tou 15:Start Time	50087	2	RW	U32	1	s	√		
47	[ESS Preheating] tou 15:End Time	50089	2	RW	U32	1	s	√		
48	[ESS Preheating] tou 15:Target Charging/ Discharging Power	50091	2	RW	S32	1000	kW	√		
49	[ESS Preheating] tou 16:Start Time	50093	2	RW	U32	1	s	√		
50	[ESS Preheating] tou 16:End Time	50095	2	RW	U32	1	s	√		
51	[ESS Preheating] tou 16:Target Charging/ Discharging Power	50097	2	RW	S32	1000	kW	√		
52	[ESS Preheating] tou 17:Start Time	50099	2	RW	U32	1	s	√		
53	[ESS Preheating] tou 17:End Time	50101	2	RW	U32	1	s	√		
54	[ESS Preheating] tou 17:Target Charging/ Discharging Power	50103	2	RW	S32	1000	kW	√		

55	[ESS Preheating] tou 18:Start Time	50105	2	RW	U32	1	s	√		
56	[ESS Preheating] tou 18:End Time	50107	2	RW	U32	1	s	√		
57	[ESS Preheating] tou 18:Target Charging/ Discharging Power	50109	2	RW	S32	1000	kW	√		
58	[ESS Preheating] tou 19:Start Time	50111	2	RW	U32	1	s	√		
59	[ESS Preheating] tou 19:End Time	50113	2	RW	U32	1	s	√		
60	[ESS Preheating] tou 19:Target Charging/ Discharging Power	50115	2	RW	S32	1000	kW	√		
61	[ESS Preheating] tou 20:Start Time	50117	2	RW	U32	1	s	√		
62	[ESS Preheating] tou 20:End Time	50119	2	RW	U32	1	s	√		
63	[ESS Preheating] tou 20:Target Charging/ Discharging Power	50121	2	RW	S32	1000	kW	√		
64	[ESS Preheating] tou 21:Start Time	50123	2	RW	U32	1	s	√		
65	[ESS Preheating] tou 21:End Time	50125	2	RW	U32	1	s	√		
66	[ESS Preheating] tou 21:Target Charging/ Discharging Power	50127	2	RW	S32	1000	kW	√		
67	[ESS Preheating] tou 22:Start Time	50129	2	RW	U32	1	s	√		
68	[ESS Preheating] tou 22:End Time	50131	2	RW	U32	1	s	√		
69	[ESS Preheating] tou 22:Target Charging/ Discharging Power	50133	2	RW	S32	1000	kW	√		
70	[ESS Preheating] tou 23:Start Time	50135	2	RW	U32	1	s	√		
71	[ESS Preheating] tou 23:End Time	50137	2	RW	U32	1	s	√		
72	[ESS Preheating] tou 23:Target Charging/ Discharging Power	50139	2	RW	S32	1000	kW	√		

73	[ESS Preheating] tou 24:Start Time	50141	2	RW	U32	1	s	√		
74	[ESS Preheating] tou 24:End Time	50143	2	RW	U32	1	s	√		
75	[ESS Preheating] tou 24:Target Charging/ Discharging Power	50145	2	RW	S32	1000	kW	√		
76	[ESS Preheating] tou 25:Start Time	50147	2	RW	U32	1	s	√		
77	[ESS Preheating] tou 25:End Time	50149	2	RW	U32	1	s	√		
78	[ESS Preheating] tou 25:Target Charging/ Discharging Power	50151	2	RW	S32	1000	kW	√		
79	[ESS Preheating] tou 26:Start Time	50153	2	RW	U32	1	s	√		
80	[ESS Preheating] tou 26:End Time	50155	2	RW	U32	1	s	√		
81	[ESS Preheating] tou 26:Target Charging/ Discharging Power	50157	2	RW	S32	1000	kW	√		
82	[ESS Preheating] tou 27:Start Time	50159	2	RW	U32	1	s	√		
83	[ESS Preheating] tou 27:End Time	50161	2	RW	U32	1	s	√		
84	[ESS Preheating] tou 27:Target Charging/ Discharging Power	50163	2	RW	S32	1000	kW	√		
85	[ESS Preheating] tou 28:Start Time	50165	2	RW	U32	1	s	√		
86	[ESS Preheating] tou 28:End Time	50167	2	RW	U32	1	s	√		
87	[ESS Preheating] tou 28:Target Charging/ Discharging Power	50169	2	RW	S32	1000	kW	√		
88	[ESS Preheating] tou 29:Start Time	50171	2	RW	U32	1	s	√		
89	[ESS Preheating] tou 29:End Time	50173	2	RW	U32	1	s	√		
90	[ESS Preheating] tou 29:Target Charging/ Discharging Power	50175	2	RW	S32	1000	kW	√		

91	[ESS Preheating] tou 30:Start Time	50177	2	RW	U32	1	s	√		
92	[ESS Preheating] tou 30:End Time	50179	2	RW	U32	1	s	√		
93	[ESS Preheating] tou 30:Target Charging/ Discharging Power	50181	2	RW	S32	1000	kW	√		
94	[ESS Preheating] Preheating reserved SOC	50183	1	RW	U16	100	%	√		Range: [0,100.0]

*Note1 : ESS Preheating registers (50000~50183), only Sigen PV M1-HYA/HYB series support.

5.2.4 Plant parameter setting others register definition (holding register)

Table 5-2-4 Plant setting parameter others register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Hybrid Inv,	PV inv.	Comment
1	Start/Stop	40000	1	WO	U16	N/A	N/A	√	√	0: Stop 1: Start
2	Reserved	40026	3	RW	N/A	N/A	N/A			
3	Remote EMS enable	40029	1	RW	U16	N/A	N/A	√	√	0: disabled 1: enabled When needed to control EMS remotely, this register needs to be enabled. When enabled, the plant's EMS work mode (30003) will switch to remote EMS.
4	Independent phase power control enable	40030	1	RW	U16	N/A	N/A	√	√	Valid only when output type is L1/L2/L3/N. To enable independent phase control, this parameter must be enabled. 0: disabled 1: enabled
5	Remote EMS control mode	40031	1	RW	U16	N/A	N/A	√	√	Mode values' definition refer to Appendix 6
6	[ESS]Backup SOC	40046	1	RW	U16	10	%	√		Range: [0,100.0]

7	[ESS] Charge Cut-Off SOC	40047	1	RW	U16	10	%	√		Range: [0,100.0]
8	[ESS] Discharge Cut-Off SOC	40048	1	RW	U16	10	%	√		Range: [0,100.0]
9	Active power regulation gradient	40049	2	RW	U32	1000	%/s	√	√	Range:[0, 5000]。Percentage of rated power adjusted per second
10	[Grid code]LVRT enable	40051	1	RW	U16	N/A	N/A	√	√	0: Disable 1: Enable
11	[Grid code]LVRT reactive power compensation factor	40052	1	RW	U16	100	N/A	√	√	Range:[0,10.0]
12	[Grid code]LVRT negative-sequence reactive power compensation factor	40053	1	RW	U16	100	N/A	√	√	Range:[0,10.0]
13	[Grid code]LVRT mode	40054	1	RW	U16	N/A	N/A	√	√	0: Reactive power compensation current, active zero-current mode 2: Zero-current mode 3: Constant current mode 4: Reactive dynamic current, active zero-current mode 5: Reactive power compensation current, active constant-current mode
14	[Grid code]LVRT grid voltage protection blocking	40055	1	RW	U16	N/A	N/A	√	√	0: Not block 1: Block
15	[Grid code]HVRT enable	40056	1	RW	U16	N/A	N/A	√	√	0: Disable 1: Enable
16	[Grid code]HVRT reactive power compensation factor	40057	1	RW	U16	100	N/A	√	√	Range:[0,10.0]
17	[Grid code]HVRT negative-sequence reactive power compensation factor	40058	1	RW	U16	100	N/A	√	√	Range:[0,10.0]
18	[Grid code]HVRT mode	40059	1	RW	U16	N/A	N/A	√	√	0: Reactive power compensation current, active zero-current mode 2: Zero-current mode

										3: Constant current mode 4: Reactive dynamic current, active hold mode 5: Reactive power compensation current, active constant-current mode
19	[Grid code]HVRT grid voltage protection blocking	40060	1	RW	U16	N/A	N/A	√	√	0: Not block 1: Block
20	[Grid code]Over-frequency derating enable	40061	1	RW	U16	N/A	N/A	√	√	0: Disable 1: Enable
21	[Grid code]Over-frequency derating power ramp rate	40062	1	RW	U16	100	%	√	√	Range:[0,100.0]
22	[Grid code]Over-frequency derating trigger frequency	40063	1	RW	U16	100	Hz	√	√	Range:[1.0*Fn, 1.2*Fn] Reference:[Grid code]Rated Frequency (Register 30276)
23	[Grid code]Over-frequency derating cut-off frequency	40064	1	RW	U16	100	Hz	√	√	Range:[1.0*Fn, 1.2*Fn] Reference:[Grid code]Rated Frequency (Register 30276)
24	[Grid code]Under-frequency power boost enable	40065	1	RW	U16	N/A	N/A	√	√	0: Disable 1: Enable
25	[Grid code]Under-frequency power boost power ramp rate	40066	1	RW	U16	100	%	√	√	Range:[0,100.0]
26	[Grid code]Under-frequency power boost trigger frequency	40067	1	RW	U16	100	Hz	√	√	Range:[0.8*Fn, 1.0*Fn] Reference:[Grid code]Rated Frequency (Register 30276)
27	[Grid code]Under-frequency power boost cut-off frequency	40068	1	RW	U16	100	Hz	√	√	Range:[0.8*Fn, 1.0*Fn] Reference:[Grid code]Rated Frequency (Register 30276)
28	Reserved	40069-40156	88	RW	N/A	N/A	N/A			
29	Grid Power Loss Lockout Alarm Clear	40159	1	WO	U16	N/A	N/A	√	√	1: Clear alarm Other values are ignored.

*Note1 : Independent phase power control registers (40008~40025, 40030), only SigenStor, Sigen Hybrid, Sigen PV M1-HYB series support.

*Note2 : Grid code registers (40051~40068), only SigenStor, Sigen Hybrid support.

5.3 Hybrid inverter(including DC Charger and ESS) running information register definition (input register)

The registers below can only be accessed with a valid Hybrid inverter's Modbus slave address (1-246), and the Modbus slave address must be configured using the MySigen APP. When using PV string related registers, please refer to the string count register(31025) value in Table 5-3 (Inverter Operating Information Registers), to ensure if the register is available.

Table 5-3 Hybrid inverter(including DC Charger and ESS) running information register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Hybrid Inv,	PV inv.	Comment
1	Model type	30500	15	RO	STRING	N/A	N/A	√	√	
2	Serial number	30515	10	RO	STRING	N/A	N/A	√	√	
3	Machine firmware version	30525	15	RO	STRING	N/A	N/A	√	√	
4	Rated active power	30540	2	RO	U32	1000	kW	√	√	
5	Max. apparent power	30542	2	RO	U32	1000	kVA	√	√	
6	Max. active power	30544	2	RO	U32	1000	kW	√	√	
7	Max. absorption power	30546	2	RO	U32	1000	kW	√		
8	Rated battery capacity	30548	2	RO	U32	100	kWh	√		
9	[ESS]Rated charge power	30550	2	RO	U32	1000	kW	√		

10	[ESS]Rated discharge power	30552	2	RO	U32	1000	kW	√		
11	Reserved	30554	12	RO	N/A	N/A	N/A			
12	[ESS]Daily charge energy	30566	2	RO	U32	100	kWh	√		
13	[ESS]Accumulated charge energy	30568	4	RO	U64	100	kWh	√		
14	[ESS]Daily discharge energy	30572	2	RO	U32	100	kWh	√		
15	[ESS]Accumulated discharge energy	30574	4	RO	U64	100	kWh	√		
16	Running state	30578	1	RO	U16	N/A	N/A	√	√	Refer to Appendix 1
17	Max.active power adjustment value	30579	2	RO	S32	1000	kW	√	√	
18	Min. active power adjustment value	30581	2	RO	S32	1000	kW	√		
19	Max. reactive power adjustment value fed to the ac terminal	30583	2	RO	U32	1000	kVar	√	√	
20	Max. reactive power adjustment value absorbed from the ac terminal	30585	2	RO	U32	1000	kVar	√	√	
21	Active power	30587	2	RO	S32	1000	kW	√	√	
22	Reactive power	30589	2	RO	S32	1000	kVar	√	√	
23	[ESS]Max. battery charge power	30591	2	RO	U32	1000	kW	√		
24	[ESS]Max. battery discharge power	30593	2	RO	U32	1000	kW	√		
25	[ESS]Available battery charge Energy	30595	2	RO	U32	100	kWh	√		
26	[ESS]Available battery discharge Energy	30597	2	RO	U32	100	kWh	√		
27	[ESS] Charge / discharge power	30599	2	RO	S32	1000	kW	√		
28	[ESS]Battery SOC	30601	1	RO	U16	10	%	√		
29	[ESS]Battery SOH	30602	1	RO	U16	10	%	√		

30	[ESS]Average cell temperature	30603	1	RO	S16	10	℃	√		
31	[ESS] Average cell voltage	30604	1	RO	U16	1000	V	√		
32	Alarm1	30605	1	RO	U16	N/A	N/A	√	√	Refer to Appendix 2
33	Alarm2	30606	1	RO	U16	N/A	N/A	√	√	Refer to Appendix 3
34	Alarm3	30607	1	RO	U16	N/A	N/A	√		Refer to Appendix 4
35	Alarm4	30608	1	RO	U16	N/A	N/A	√	√	Refer to Appendix 5
36	Alarm5	30609	1	RO	U16	N/A	N/A	√		Refer to Appendix 11
37	Reserved	30610	3	RO	N/A	N/A	N/A			
38	Active power fixed value adjustment feedback	30613	2	RO	S32	1000	kW	√	√	
39	Reactive power fixed value adjustment feedback	30615	2	RO	S32	1000	kVar	√	√	
40	Active power percentage adjustment feedback	30617	1	RO	S16	100	%	√	√	
41	Reactive power Q/S adjustment feedback	30618	1	RO	S16	100	%	√	√	
42	Power factor adjustment feedback	30619	1	RO	S16	1000	N/A	√	√	
43	[ESS]Maximum battery (cluster) temperature	30620	1	RO	S16	10	℃	√		
44	[ESS]Minimum battery (cluster) temperature	30621	1	RO	S16	10	℃	√		
45	[ESS] Maximum battery (cluster) cell voltage	30622	1	RO	U16	1000	V	√		
46	[ESS] Minimum battery (cluster) cell voltage	30623	1	RO	U16	1000	V	√		
47	Rated grid voltage	31000	1	RO	U16	10	V	√	√	
48	Rated grid frequency	31001	1	RO	U16	100	Hz	√	√	
49	Grid frequency	31002	1	RO	U16	100	Hz	√	√	

50	[PCS] Internal temperature	31003	1	RO	S16	10	℃	√	√	
51	Output type	31004	1	RO	U16	N/A	N/A	√	√	0: L/N 1: L1/L2/L3 2: L1/L2/L3/N 3: L1/L2/N
52	A-B line voltage	31005	2	RO	U32	100	V	√	√	Invalid when output type is L/N, L1/L2/N, or L1/L2/N
53	B-C line voltage	31007	2	RO	U32	100	V	√	√	
54	C-A line voltage	31009	2	RO	U32	100	V	√	√	
55	Phase A voltage	31011	2	RO	U32	100	V	√	√	When output type is L/N, refers to "Phase voltage"
56	Phase B voltage	31013	2	RO	U32	100	V	√	√	Invalid when output type is L/N, L1/L2/N, or L1/L2/N
57	Phase C voltage	31015	2	RO	U32	100	V	√	√	
58	Phase A current	31017	2	RO	S32	100	A	√	√	When output type is L/N, refers to "Phase current"
59	Phase B current	31019	2	RO	S32	100	A	√	√	Invalid when output type is L/N, L1/L2/N, or L1/L2/N
60	Phase C current	31021	2	RO	S32	100	A	√	√	
61	Power factor	31023	1	RO	S16	1000	N/A	√	√	
62	PACK count	31024	1	RO	U16	1	N/A	√		
63	PV string count	31025	1	RO	U16	1	N/A	√	√	
64	MPPT count	31026	1	RO	U16	1	N/A	√	√	
65	PV1 voltage	31027	1	RO	S16	10	V	√	√	Please refer to the PV count listed in Tabel2-1 in chapter 2, to ensure if the register is available.
66	PV1 current	31028	1	RO	S16	100	A	√	√	
67	PV2 voltage	31029	1	RO	S16	10	V	√	√	

68	PV2 current	31030	1	RO	S16	100	A	√	√	
69	PV3 voltage	31031	1	RO	S16	10	V	√	√	
70	PV3 current	31032	1	RO	S16	100	A	√	√	
71	PV4 voltage	31033	1	RO	S16	10	V	√	√	
72	PV4 current	31034	1	RO	S16	100	A	√	√	
73	PV power	31035	2	RO	S32	1000	kW	√	√	
74	Insulation resistance	31037	1	RO	U16	1000	MΩ	√	√	
75	Startup time	31038	2	RO	U32	1	s	√	√	
76	Shutdown time	31040	2	RO	U32	1	s	√	√	
77	PV5 voltage	31042	1	RO	S16	10	V	√	√	Please refer to the PV count listed in Tabel2-1 in chapter 2, to ensure if the register is available.
78	PV5 current	31043	1	RO	S16	100	A	√	√	
79	PV6 voltage	31044	1	RO	S16	10	V	√	√	
80	PV6 current	31045	1	RO	S16	100	A	√	√	
81	PV7 voltage	31046	1	RO	S16	10	V	√	√	
82	PV7 current	31047	1	RO	S16	100	A	√	√	
83	PV8 voltage	31048	1	RO	S16	10	V	√	√	
84	PV8 current	31049	1	RO	S16	100	A	√	√	
85	PV8 current	31050	1	RO	S16	10	V	√	√	
86	PV9 current	31051	1	RO	S16	100	A	√	√	
87	PV10 voltage	31052	1	RO	S16	10	V	√	√	

88	PV10 current	31053	1	RO	S16	100	A	√	√	
89	PV11 voltage	31054	1	RO	S16	10	V	√	√	
90	PV11 current	31055	1	RO	S16	100	A	√	√	
91	PV12 voltage	31056	1	RO	S16	10	V	√	√	
92	PV12 current	31057	1	RO	S16	100	A	√	√	
93	PV13 voltage	31058	1	RO	S16	10	V	√	√	
94	PV13 current	31059	1	RO	S16	100	A	√	√	
95	PV14 voltage	31060	1	RO	S16	10	V	√	√	
96	PV14 current	31061	1	RO	S16	100	A	√	√	
97	PV15 voltage	31062	1	RO	S16	10	V	√	√	
98	PV15 current	31063	1	RO	S16	100	A	√	√	
99	PV16 voltage	31064	1	RO	S16	10	V	√	√	
100	PV16 current	31065	1	RO	S16	100	A	√	√	
101	PV17 voltage	31066	1	RO	S16	10	V		√	
102	PV17 current	31067	1	RO	S16	100	A		√	
103	PV18 voltage	31068	1	RO	S16	10	V		√	
104	PV18 current	31069	1	RO	S16	100	A		√	
105	PV19 voltage	31070	1	RO	S16	10	V		√	
106	PV19 current	31071	1	RO	S16	100	A		√	
107	PV20 voltage	31072	1	RO	S16	10	V		√	
108	PV20 current	31073	1	RO	S16	100	A		√	

109	PV21 voltage	31074	1	RO	S16	10	V		√	
110	PV21 current	31075	1	RO	S16	100	A		√	
111	PV22 voltage	31076	1	RO	S16	10	V		√	
112	PV22 current	31077	1	RO	S16	100	A		√	
113	PV23 voltage	31078	1	RO	S16	10	V		√	
114	PV23 current	31079	1	RO	S16	100	A		√	
115	PV24 voltage	31080	1	RO	S16	10	V		√	
116	PV24 current	31081	1	RO	S16	100	A		√	
117	PV25 voltage	31082	1	RO	S16	10	V		√	
118	PV25 current	31083	1	RO	S16	100	A		√	
119	PV26 voltage	31084	1	RO	S16	10	V		√	
120	PV26 current	31085	1	RO	S16	100	A		√	
121	PV27 voltage	31086	1	RO	S16	10	V		√	
122	PV27 current	31087	1	RO	S16	100	A		√	
123	PV28 voltage	31088	1	RO	S16	10	V		√	
124	PV28 current	31089	1	RO	S16	100	A		√	
125	PV29 voltage	31090	1	RO	S16	10	V		√	
126	PV29 current	31091	1	RO	S16	100	A		√	
127	PV30 voltage	31092	1	RO	S16	10	V		√	
128	PV30 current	31093	1	RO	S16	100	A		√	
129	PV31 voltage	31094	1	RO	S16	10	V		√	

130	PV31 current	31095	1	RO	S16	100	A		√	
131	PV32 voltage	31096	1	RO	S16	10	V		√	
132	PV32 current	31097	1	RO	S16	100	A		√	
133	PV33 voltage	31098	1	RO	S16	10	V		√	
134	PV33 current	31099	1	RO	S16	100	A		√	
135	PV34 voltage	31100	1	RO	S16	10	V		√	
136	PV34 current	31101	1	RO	S16	100	A		√	
137	PV35 voltage	31102	1	RO	S16	10	V		√	
138	PV35 current	31103	1	RO	S16	100	A		√	
139	PV36 voltage	31104	1	RO	S16	10	V		√	
140	PV36 current	31105	1	RO	S16	100	A		√	
141	[DC Charger] Vehicle battery voltage	31500	1	RO	U16	10	V	√		
142	[DC Charger] Charging current	31501	1	RO	U16	10	A	√		
143	[DC Charger] Output power	31502	2	RO	S32	1000	kW	√		
144	[DC Charger] Vehicle SOC	31504	1	RO	U16	10	%	√		
145	[DC Charger] Current charging capacity	31505	2	RO	U32	100	kWh	√		Single time
146	[DC Charger] Current charging duration	31507	2	RO	U32	1	s	√		Single time
147	PV daily generation	31509	2	RO	U32	100	kWh	√	√	
148	PV total generation	31511	2	RO	U32	100	kWh	√	√	
149	[DC Charger] Running state	31513	1	RO	U16	N/A	N/A	√		Refer to Appendix 14.

150	[DC Charger] Discharging current	31514	1	RO	U16	10	A	√		
151	[DC Charger]Current discharging capacity	31515	2	RO	U32	100	kWh	√		Single time
152	[DC Charger]Current discharging duration	31517	2	RO	U32	1	s	√		Single time
153	[DC Charger]Total charging capacity	31519	2	RO	U32	100	kWh	√		
154	[DC Charger]Total Discharging capacity	31521	2	RO	U32	100	kWh	√		
155	[DC Charger]Rated charging power	31523	2	RO	U32	1000	kW	√		
156	[DC Charger]Rated discharging power	31525	2	RO	U32	1000	kW	√		

*Note : DC Charger only SigenStor, Sigen Hybrid support.

5.4 Hybrid inverter(including DC Charger and ESS) parameter setting register definition (holding register)

The registers below can only be accessed with a valid Hybrid inverter's Modbus slave address (1-246), and the Modbus slave address must be configured using the MySigen APP.

Table 5-4 Hybrid inverter(including DC Charger and ESS) parameter setting register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Hybrid Inv,	PV inv.	Comment
1	Start/Stop	40500	1	WO	U16	N/A	N/A	√	√	0: Stop 1: Start
2	Reserved	40501	1	RW	U16	N/A	N/A			
3	[DC Charger] Start/Stop	41000	1	WO	U16	N/A	N/A	√		0: Start 1: Stop
4	Reserved	41001	1	WO	U16	N/A	N/A			
5	[DC Charger] max charging power limit	41002	2	RW	U32	1000	kW	√		[0, Rated charging power].

6	[DC Charger] max discharging power limit	41004	2	RW	U32	1000	kW	√		[0, Rated discharging power].
7	Reserved	41500	1	RW	U16	N/A	N/A		√	
8	Active power fixed value adjustment	41501	2	RW	S32	1000	kW		√	
9	Reactive power fixed value adjustment	41503	2	RW	S32	1000	kVar		√	
10	Active power percentage adjustment	41505	1	RW	S16	100	%		√	
11	Reactive power Q/S adjustment	41506	1	RW	S16	100	%		√	
12	Power factor adjustment	41507	1	RW	S16	1000	N/A		√	

5.5 AC-Charger running information register definition (input register)

The registers below can only be accessed with a valid AC-Charger's Modbus slave address (1-246), and the Modbus slave address must be configured using the MySigen APP. And are only applicable for "EVAC" devices.

Table 5-5 AC-Charger running information register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Comment
1	System state	32000	1	RO	U16	N/A	N/A	System states according to IEC61851-1 definition. Refer to Appendix 7.
2	Total energy consumed	32001	2	RO	U32	100	kWh	
3	Charging power	32003	2	RO	S32	1000	kW	

4	Rated power	32005	2	RO	U32	1000	kW	
5	Rated current	32007	2	RO	S32	100	A	
6	Rated voltage	32009	1	RO	U16	10	V	
7	AC-Charger input breaker rated current	32010	2	RO	S32	100	A	
8	Alarm1	32012	1	RO	U16	N/A	N/A	Refer to Appendix 8
9	Alarm2	32013	1	RO	U16	N/A	N/A	Refer to Appendix 9
10	Alarm3	32014	1	RO	U16	N/A	N/A	Refer to Appendix 10

5.6 AC-Charger parameter setting register definition (holding register)

The registers below can only be accessed with a valid AC-Charger's modbus slave address (1-246), and the Modbus slave address must be configured using the MySigen APP. And are only applicable for "EVAC" devices.

Table 5-6 AC-Charger parameter setting register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Comment
1	Start/Stop	42000	1	WO	U16	N/A	N/A	0: Start 1: Stop
2	Charger output current	42001	2	RW	U32	100	N/A	[6, X] X is the smaller value between the rated current and the AC-Charger input breaker rated current.
3	Reserved	42003	1	WO	U16	N/A	N/A	

5.7 PSS running information register definition (input register)

The registers below can only be accessed with a valid PSS Modbus slave address (1-246).

Table 5-7 PSS running information register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Comment
1	Model type	32500	15	RO	STRING	N/A	N/A	
2	Serial number	32515	10	RO	STRING	N/A	N/A	
3	Communication status	32525	1	RO	U16	N/A	N/A	0: Loading 1: Offline 2: Online
4	Teleindication1	32526	1	RO	BIT16	N/A	N/A	Refer to Appendix 15
5	Teleindication2	32527	1	RO	BIT16	N/A	N/A	Refer to Appendix 16
6	Teleindication3	32528	1	RO	BIT16	N/A	N/A	Refer to Appendix 17
7	Teleindication4	32529	1	RO	BIT16	N/A	N/A	Refer to Appendix 18
8	Teleindication5	32530	1	RO	BIT16	N/A	N/A	Refer to Appendix 19
9	[MV] Phase A current	32531	2	RO	S32	100	A	
10	[MV] Phase B current	32533	2	RO	S32	100	A	
11	[MV] phase C current	32535	2	RO	S32	100	A	
12	[MV] zero sequence current	32537	2	RO	S32	100	A	
13	[MV] frequency	32539	1	RO	U16	100	Hz	
14	[MV] temperature	32540	1	RO	S16	10	℃	
15	[MV] humidity	32541	1	RO	U16	10	RH	
16	[MV] G1 cable L1 phase temperature	32542	1	RO	S16	10	℃	
17	[MV] G1 cable L2 phase temperature	32543	1	RO	S16	10	℃	
18	[MV] G1 cable L3 phase temperature	32544	1	RO	S16	10	℃	
19	[MV] G2 cable L1 phase temperature	32545	1	RO	S16	10	℃	
20	[MV] G2 cable L2 phase temperature	32546	1	RO	S16	10	℃	
21	[MV] G2 cable L3 phase temperature	32547	1	RO	S16	10	℃	
22	[MV] G3 cable L1 phase temperature	32548	1	RO	S16	10	℃	
23	[MV] G3 cable L2 phase temperature	32549	1	RO	S16	10	℃	
24	[MV] G3 cable L3 phase temperature	32550	1	RO	S16	10	℃	
25	[LA] temperature	32551	1	RO	S16	10	℃	
26	[LA] humidity	32552	1	RO	U16	10	RH	
27	[LA] phase A voltage	32553	2	RO	U32	100	V	
28	[LA] phase B voltage	32555	2	RO	U32	100	V	
29	[LA] phase C voltage	32557	2	RO	U32	100	V	

30	[LA] AB line voltage	32559	2	RO	U32	100	V	
31	[LA] BC line voltage	32561	2	RO	U32	100	V	
32	[LA] CA line voltage	32563	2	RO	U32	100	V	
33	[LA] phase A current	32565	2	RO	S32	100	A	
34	[LA] phase B current	32567	2	RO	S32	100	A	
35	[LA] phase C current	32569	2	RO	S32	100	A	
36	[LA] phase A active power	32571	2	RO	S32	1000	kW	
37	[LA] phase B active power	32573	2	RO	S32	1000	kW	
38	[LA] phase C active power	32575	2	RO	S32	1000	kW	
39	[LA] total active power	32577	2	RO	S32	1000	kW	
40	[LA] phase A reactive power	32579	2	RO	S32	1000	kVar	
41	[LA] phase B reactive power	32581	2	RO	S32	1000	kVar	
42	[LA] phase C reactive power	32583	2	RO	S32	1000	kVar	
43	[LA] total reactive power	32585	2	RO	S32	1000	kVar	
44	[LA] phase A apparent power	32587	2	RO	U32	1000	kVA	
45	[LA] phase B apparent power	32589	2	RO	U32	1000	kVA	
46	[LA] phase C apparent power	32591	2	RO	U32	1000	kVA	
47	[LA] total apparent power	32593	2	RO	U32	1000	kVA	
48	[LA] phase A power factor	32595	1	RO	S16	1000	N/A	
49	[LA] phase B power factor	32596	1	RO	S16	1000	N/A	
50	[LA] phase C power factor	32597	1	RO	S16	1000	N/A	
51	[LA] total power factor	32598	1	RO	S16	1000	N/A	
52	[LA] frequency	32599	1	RO	U16	100	Hz	
53	[LA] forward active energy	32600	4	RO	U64	100	kWh	
54	[LA] reverse active energy	32604	4	RO	U64	100	kWh	
55	[LA] forward reactive energy	32608	4	RO	U64	100	kvarh	

56	[LA] reverse reactive energy	32612	4	RO	U64	100	kvarh	
57	[LB] temperature	32616	1	RO	S16	10	℃	
58	[LB] humidity	32617	1	RO	U16	10	RH	
59	[LB] phase A voltage	32618	2	RO	U32	100	V	
60	[LB] phase B voltage	32620	2	RO	U32	100	V	
61	[LB] phase C voltage	32622	2	RO	U32	100	V	
62	[LB] AB line voltage	32624	2	RO	U32	100	V	
63	[LB] BC line voltage	32626	2	RO	U32	100	V	
64	[LB] CA line voltage	32628	2	RO	U32	100	V	
65	[LB] phase A current	32630	2	RO	S32	100	A	
66	[LB] phase B current	32632	2	RO	S32	100	A	
67	[LB] phase C current	32634	2	RO	S32	100	A	
68	[LB] phase A active power	32636	2	RO	S32	1000	kW	
69	[LB] phase B active power	32638	2	RO	S32	1000	kW	
70	[LB] phase C active power	32640	2	RO	S32	1000	kW	
71	[LB] total active power	32642	2	RO	S32	1000	kW	
72	[LB] phase A reactive power	32644	2	RO	S32	1000	kVar	
73	[LB] phase B reactive power	32646	2	RO	S32	1000	kVar	
74	[LB] phase C reactive power	32648	2	RO	S32	1000	kVar	
75	[LB] total reactive power	32650	2	RO	S32	1000	kVar	
76	[LB] phase A apparent power	32652	2	RO	U32	1000	kVA	
77	[LB] phase B apparent power	32654	2	RO	U32	1000	kVA	
78	[LB] phase C apparent power	32656	2	RO	U32	1000	kVA	
79	[LB] total apparent power	32658	2	RO	U32	1000	kVA	
80	[LB] phase A power factor	32660	1	RO	S16	1000	N/A	
81	[LB] phase B power factor	32661	1	RO	S16	1000	N/A	
82	[LB] phase C power factor	32662	1	RO	S16	1000	N/A	
83	[LB] total power factor	32663	1	RO	S16	1000	N/A	

84	[LB] frequency	32664	1	RO	U16	100	Hz	
85	[LB] forward active energy	32665	4	RO	U64	100	kWh	
86	[LB] reverse active energy	32669	4	RO	U64	100	kWh	
87	[LB] forward reactive energy	32673	4	RO	U64	100	kvarh	
88	[LB] reverse reactive energy	32677	4	RO	U64	100	kvarh	
89	[transformer] oil surface temperature	32681	1	RO	S16	10	℃	
90	[transformer] winding temperature	32682	1	RO	S16	10	℃	
91	[distribution cabinet] temperature	32683	1	RO	S16	10	℃	
92	[distribution cabinet] humidity	32684	1	RO	U16	10	RH	
93	[distribution cabinet] phase A voltage	32685	2	RO	U32	100	V	
94	[distribution cabinet] phase B voltage	32687	2	RO	U32	100	V	
95	[distribution cabinet] phase C voltage	32689	2	RO	U32	100	V	
96	[distribution cabinet] AB line voltage	32691	2	RO	U32	100	V	
97	[distribution cabinet] BC line voltage	32693	2	RO	U32	100	V	
98	[distribution cabinet] CA line voltage	32695	2	RO	U32	100	V	
99	[distribution cabinet] phase A current	32697	2	RO	S32	100	A	
100	[distribution cabinet] phase B current	32699	2	RO	S32	100	A	
101	[distribution cabinet] phase C current	32701	2	RO	S32	100	A	
102	[distribution cabinet] phase A active power	32703	2	RO	S32	1000	kW	
103	[distribution cabinet] phase B active power	32705	2	RO	S32	1000	kW	
104	[distribution cabinet] phase C active power	32707	2	RO	S32	1000	kW	
105	[distribution cabinet] total active power	32709	2	RO	S32	1000	kW	

106	[distribution cabinet] phase A reactive power	32711	2	RO	S32	1000	kVar	
107	[distribution cabinet] phase B reactive power	32713	2	RO	S32	1000	kVar	
108	[distribution cabinet] phase C reactive power	32715	2	RO	S32	1000	kVar	
109	[distribution cabinet] total reactive power	32717	2	RO	S32	1000	kVar	
110	[distribution cabinet] phase A apparent power	32719	2	RO	U32	1000	kVA	
111	[distribution cabinet] phase B apparent power	32721	2	RO	U32	1000	kVA	
112	[distribution cabinet] phase C apparent power	32723	2	RO	U32	1000	kVA	
113	[distribution cabinet] total apparent power	32725	2	RO	U32	1000	kVA	
114	[distribution cabinet] phase A power factor	32727	1	RO	S16	1000	N/A	
115	[distribution cabinet] phase B power factor	32728	1	RO	S16	1000	N/A	
116	[distribution cabinet] phase C power factor	32729	1	RO	S16	1000	N/A	
117	[distribution cabinet] total power factor	32730	1	RO	S16	1000	N/A	
118	[distribution cabinet] frequency	32731	1	RO	U16	100	Hz	
119	[distribution cabinet] forward active energy	32732	4	RO	U64	100	kWh	
120	[distribution cabinet] reverse active energy	32736	4	RO	U64	100	kWh	
121	[distribution cabinet] forward reactive energy	32740	4	RO	U64	100	kvarh	
122	[distribution cabinet] reverse reactive energy	32744	4	RO	U64	100	kvarh	

5.8 PSS parameter setting register definition (holding register)

The registers below can only be accessed with a valid PSS modbus slave address (1-246).

Table 5-8 PSS parameter setting register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Comment
1	Medium voltage cabinet G3 circuit breaker switch-on	42500	1	WO	U16	N/A	N/A	0: No action 1: switch-on
2	Medium voltage cabinet G3 circuit breaker switch-off	42501	1	WO	U16	N/A	N/A	0: No action 1: switch-off
3	LA low voltage cabinet circuit breaker switch-on	42502	1	WO	U16	N/A	N/A	0: No action 1: switch-on
4	LA low voltage cabinet circuit breaker switch-off	42503	1	WO	U16	N/A	N/A	0: No action 1: switch-off
5	LB low voltage cabinet circuit breaker switch-on	42504	1	WO	U16	N/A	N/A	0: No action 1: switch-on
6	LB low voltage cabinet circuit breaker switch-off	42505	1	WO	U16	N/A	N/A	0: No action 1: switch-off

5.9 PID running information register definition (input register)

The registers below can only be accessed with a valid PID Modbus slave address (1-246).

Table 5-9 PID running information register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Comment
1	Model type	33000	15	RO	STRING	N/A	N/A	
2	Serial number	33015	10	RO	STRING	N/A	N/A	
3	Machine firmware	33025	15	RO	STRING	N/A	N/A	

	version							
4	Communication status	33040	1	RO	U16	N/A	N/A	0: Loading 1: Offline 2: Online
5	Running status	33041	1	RO	U16	N/A	N/A	0: Idle 1: Fault 2: IMD detection 3: PID compensation
6	AB line voltage	33042	2	RO	U32	100	V	
7	BC line voltage	33044	2	RO	U32	100	V	
8	CA line voltage	33046	2	RO	U32	100	V	
9	Phase A voltage	33048	2	RO	U32	100	V	
10	Phase B voltage	33050	2	RO	U32	100	V	
11	Phase C voltage	33052	2	RO	U32	100	V	
12	Grid frequency	33054	1	RO	U16	100	Hz	
13	Output voltage	33055	1	RO	U16	10	V	
14	Bus voltage	33056	1	RO	U16	10	V	
15	Inverter voltage	33057	2	RO	U32	100	V	
16	Inverter current	33059	1	RO	U16	100	mA	
17	Output current	33060	1	RO	S16	100	mA	
18	Internal temperature 1	33061	1	RO	S16	10	℃	
19	Internal temperature 2	33062	1	RO	S16	10	℃	
20	Internal temperature 3	33063	1	RO	S16	10	℃	
21	Internal temperature 4	33064	1	RO	S16	10	℃	
22	Internal temperature 5	33065	1	RO	S16	10	℃	
23	Alarm 1	33066	1	RO	U16	N/A	N/A	Refer to Appendix 20
24	Alarm 2	33067	1	RO	U16	N/A	N/A	Refer to Appendix 21

5.10 PID parameter setting register definition (holding register)

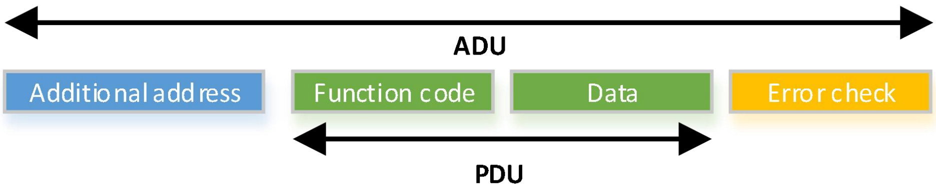
The registers below can only be accessed with a valid PID modbus slave address (1-246).

Table 5-10 PID parameter setting register definition

No.	Name	Add.	QTY	Perm.	Data Type	Gain	Unit	Comment
1	Start/stop	43000	1	WO	U16	N/A	N/A	0: Stop 1: Start

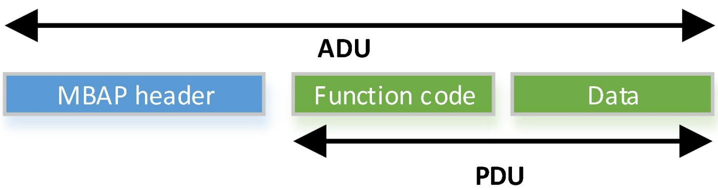
6. Modbus Protocol Command Overview

(1) MODBUS-RTU frame format



Filed	Length(Bytes)	Description
Slave Address	1	Customized by user (1~247)
PDU	X	Described in chapter 6.1
Error Check	2	Crc16 check. It is worth pointing out that the byte order of CRC16 is the little-end mode

(2) MODBUS-TCP frame format



Filed	Length(Bytes)	Description
Transmission identifier	2	Matching identifier between a request frame and a response frame
Protocol type	2	0 = Modbus protocol
Data length	2	Follow-up data length
Slave Address	1	Customized by user (1~247)

6.1 Function code

Index	Function code	Description
1	0x03	Read Holding Register
2	0x04	Read Input Register
3	0x06	Write a single Register

4	0x10	Write multiple Registers

6.1.1 Read Holding Register

Request

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Function code	1 Byte	0x03
Starting address	2 Bytes	0x0000~0xFFFF
Quantity of registers	2 Bytes	1~124

Response

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Function code	1 Byte	0x03
Byte count	1 Byte	2 x N
Register value	2 x N Bytes	N=Quantity of Registers

Error

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Error code	1 Byte	0x83
Exception code	1 Byte	01 or 02 or 03 or 04

Example PDU: The following example contains only the slave address and the Protocol Data Unit (PDU). If using Modbus RTU mode, a CRC16 should be added at the end; if using Modbus TCP mode, an MBAP header should be added at the beginning.

Example Meaning: Read the *Rated active power* register of the hybrid inverter with Modbus slave address 1.

Host Query Command: 01 03 77 4C 00 02

Slave Normal Response: 01 03 04 00 00 61 A8

Slave Exception Response: 01 83 04

6.1.2 Read Input Register

Request

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Function code	1 Byte	0x04
Starting address	2 Bytes	0x0000~0xFFFF
Quantity of registers	2 Bytes	1~124

Response

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Function code	1 Byte	0x04
Byte count	1 Byte	2 x N
Register value	2 x N Bytes	N=Quantity of Registers

Error

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Error code	1 Byte	0x84
Exception code	1 Byte	01 or 02 or 03 or 04

Example PDU: The following example contains only the slave address and the Protocol Data Unit (PDU). If using Modbus RTU mode, a CRC16 should be added at the end; if using Modbus TCP mode, an MBAP header should be added at the beginning.

Example Meaning: Read the *Active power fixed adjustment target value* register of a power plant with Modbus slave address 247.

Host Query Command: F7 04 9C 41 00 02

Slave Normal Response: F7 04 04 00 00 61 A8

Slave Exception Response: F7 84 04

6.1.3 Write a single Register

Request

Filed	Length(Bytes)	Description
Slave address	1 Byte	0~247
Function code	1 Byte	0x06
Register address	2 Bytes	0x0000~0xFFFF
Register value	2 Bytes	0x0000~0xFFFF

Response

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Function code	1 Byte	0x06
Register address	2 Bytes	0x0000~0xFFFF
Register value	2 Bytes	0x0000~0xFFFF

Error

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Error code	1 Byte	0x86
Exception code	1 Byte	01 or 02 or 03 or 04

Example PDU: The following example contains only the slave address and the Protocol Data Unit (PDU). If using Modbus RTU mode, a CRC16 should be added at the end; if using Modbus TCP mode, an MBAP header should be added at the beginning.

Example Meaning: Write the *Grid code* register of the hybrid inverter with Modbus slave address 1.

Host Query Command: 01 06 9E 34 00 01

Slave Normal Response:01 06 9E 34 00 01

Slave Exception Response: 01 86 04

6.1.4 Write multiple Registers

Request

Filed	Length(Bytes)	Description
Slave address	1 Byte	0~247
Function code	1 Byte	0x10
Starting address	2 Bytes	0x0000~0xFFFF
Quantity of registers	2 Bytes	1~123
Byte count	1 Byte	2 x N
Registers value	2 x N Bytes	N=Quantity of Registers

Response

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Function code	1 Byte	0x10
Starting address	2 Bytes	0x0000~0xFFFF
Quantity of registers	2 Bytes	1~123

Error

Filed	Length(Bytes)	Description
Slave address	1 Byte	1~247
Error code	1 Byte	0x90
Exception code	1 Byte	01 or 02 or 03 or 04

Example PDU: The following example contains only the slave address and the Protocol Data Unit (PDU). If using Modbus RTU mode, a CRC16 should be added at the end; if using Modbus TCP mode, an MBAP header should be added at the beginning.

Example Meaning: Write the *Active power fixed adjustment target value* register of a power plant with Modbus slave address 247.

Host Query Command: F7 10 9C 41 00 02 04 00 00 61 A8

Slave Normal Response: F7 10 9C 41 00 02

Slave Exception Response: F7 90 04

6.2 Exception code

Code	Name	Meaning
0x01	ILLEGAL FUNCTION	The function code received in the query is not an allowable action for the server (or slave). This may be because the function code is only applicable to newer devices, and was not implemented in the unit selected. It could also indicate that the server (or slave) is in the wrong state to process a request of this type, for example because it is unconfigured and is being asked to return register values.
0x02	ILLEGAL DATA ADDRESS	The data address received in the query is not an allowable address for the server (or slave). More specifically, the combination of reference number and transfer length is invalid.
0x03	ILLEGAL DATA VALUE	A value contained in the query data field is not an allowable value for server (or slave). This indicates a fault in the structure of the remainder of a complex request, such as that the implied length is incorrect. It specifically does NOT mean that a data item submitted for storage in a register has a value outside the expectation of the application program, since the MODBUS protocol is unaware of the significance of any particular value of any particular register.
0x04	SLAVE DEVICE FAILURE	An unrecoverable error occurred while the server (or slave) was attempting to perform the requested action.

Appendix 1 Running state

Running State	Value
Standby	0x00
Running	0x01
Fault	0x02
Shutdown	0x03
Environmental Abnormality	0x07

Appendix 2 PCS alarm code -- 1

Alarm Description	Bit
Software version mismatch	0
Low insulation resistance	1
Over-temperature	2

Equipment fault	3
System grounding fault	4
PV string over-voltage	5
PV string reversely connected	6
PV string back-filling	7
AFCI fault	8
Grid power outage	9
Grid over-voltage	10
Grid under-voltage	11
Grid over-frequency	12
Grid under-frequency	13
Grid voltage imbalance	14
DC component of output current out of limit	15

Appendix 3 PCS alarm code -- 2

Alarm Description	Bit
Leak current out of limit	0
Communication abnormal	1
System internal protection	2
AFCI self-checking circuit fault	3
Off-grid protection	4
Manual operation protection	5

Abnormal phase sequence	7
Short circuit to PE	8
Soft start failure	9
Not defined	Not defined

Appendix 4 ESS alarm code

Alarm Description	Bit
Software version mismatch	0
The energy storage module has low insulation resistance to ground	1
The temperature is too high	2
Equipment fault	3
Under-temperature	4
Internal protection	5
Thermal runaway	6
Not defined	Not defined

Appendix 5 Gateway alarm code

Alarm Description	Bit
Software version mismatch	0
The temperature is too high	1

Equipment fault	2
Excessive leakage current in off-grid output	3
N line grounding fault	4
Abnormal phase sequence of grid wiring	5
Abnormal phase sequence of inverter wiring	6
Grid phase loss	7
Not defined	Not defined

Appendix 6 Remote EMS control mode

Remote EMS control mode	Value
PCS remote control	0x00
Standby	0x01
Maximum self-consumption	0x02
Command charging (consume grid power first)	0x03
Command charging (consume PV power first)	0x04
Command discharging (output from PV first)	0x05
Command discharging (output from ESS first)	0x06
Reserved	0x07
V2G	0x08

Appendix 7 AC-Charger system state

System State	Value
System innit	0x00
A1/A2	0x01
B1	0x02
B2	0x03
C1	0x04
C2	0x05
F	0x06
E	0x07

Appendix 8 AC-Charger alarm code -- 1

Alarm Description	Bit
Grid overvoltage	0
Grid undervoltage	1
Overload	2
Short circuit	3
Charging output overcurrent	4
Leak current out of limit	5
Grounding fault	6
Abnormal phase sequence of grid wiring	7

PEN Fault	8
Not defined	Not defined

Appendix 9 AC-Charger alarm code -- 2

Alarm Description	Bit
Leak current detection circuit fault	0
Relay stuck	1
Pilot circuit fault	2
Auxiliary power supply module fault	3
Electric lock fault	4
Lamp panel communication fault	5
Not defined	Not defined

Appendix 10 AC-Charger alarm code -- 3

Alarm Description	Bit
Too high internal temperature	0
Charging cable fault	1
Meter communication fault	2
Not defined	Not defined

Appendix 11 DC-Charger alarm code

Alarm Description	Bit
Software version mismatch	0
Low insulation resistance to ground	1
Over-temperature	2
Equipment fault	3
Charging fault	4
Equipment protection	5
Not defined	Not defined

Appendix 12 Plant alarm -- 1

Alarm Description	Bit
Gateway communication abnormal	0
Meter communication abnormal	1
AC power sensor communication abnormal	2
Hard protection against grid-feed power limit exceedance	6
Generator failure to start	8
CLS fault	10
Not defined	Not defined

Appendix 13 Plant alarm -- 2

Alarm Description	Bit
OVGR fault	0
RPR fault	1
Not defined	Not defined

Appendix 14 DC-Charger running state

Running State	Value
Idle	0x00
Occupied (Charging Gun plugged in but not detected)	0x01
Preparing (Establishing communication)	0x02
Charging	0x03
Fault	0x04
Scheduled	0x05
Ended	0x06
Unavailable (Under maintenance)	0x07
Discharging	0x08
Alarm	0x09
Preparing (Insulation detection in progress)	0x0A

Appendix 15 PSS teleindication -- 1

Teleindication Description	Bit
Measurement & control unit general trip	0
Measurement & control unit general alarm	1
Transformer heavy gas trip	2
Transformer light gas alarm	3
Transformer pressure relief trip	4
Transformer low oil level alarm	5
Transformer high oil level alarm	6
Transformer oil high temperature alarm	7
Transformer oil over-temperature trip	8
Transformer winding high temperature alarm	9
Transformer winding over-temperature trip	10
Low voltage room dual smoke sensor trip	11
Medium voltage room dual smoke sensor trip	12
Low voltage room maintenance door open trip	13
Transformer room door open trip	14
LA low voltage cabinet over-temperature trip	15

Appendix 16 PSS teleindication -- 2

Teleindication Description	Bit

LB low voltage cabinet over-temperature trip	0
Medium voltage room over-temperature trip	1
Medium voltage cabinet insulation gas low pressure alarm	2
Emergency stop	3
Medium voltage room air conditioner/heat exchanger fault	4
LA Low voltage room heat exchanger fault	5
LB Low voltage room heat exchanger fault	6
Low voltage room smoke sensor alarm	7
Medium voltage room smoke sensor alarm	8
Medium voltage cabinet G3 circuit breaker switch-off failure	9
LA low voltage cabinet SPD fault	10
LB low voltage cabinet SPD fault	11
Distribution cabinet SPD fault	12
LA low voltage cabinet over-temperature alarm	13
LB low voltage cabinet over-temperature alarm	14
Medium voltage room over-temperature alarm	15

Appendix 17 PSS teleindication -- 3

Teleindication Description	Bit
LA low voltage cabinet IMD pre-alarm	0
LA low voltage cabinet IMD alarm	1

LB low voltage cabinet IMD pre-alarm	2
LB low voltage cabinet IMD alarm	3
UPS fault alarm	4
LA low voltage cabinet circuit breaker switch-on	5
LA low voltage cabinet circuit breaker switch-off	6
LA low voltage cabinet circuit breaker fault trip	7
LA low voltage cabinet circuit breaker remote control	8
LB low voltage cabinet circuit breaker switch-on	9
LB low voltage cabinet circuit breaker switch-off	10
LB low voltage cabinet circuit breaker fault trip	11
LB low voltage cabinet circuit breaker remote control	12
Medium voltage cabinet G3 circuit breaker switch-on	13
Medium voltage cabinet G3 circuit breaker switch-off	14
Medium voltage cabinet G3 disconnector switch switch-on	15

Appendix 18 PSS teleindication -- 4

Teleindication Description	Bit
Medium voltage cabinet G3 disconnector switch switch-off	0
Medium voltage cabinet G3 earthing switch switch-on	1
Medium voltage cabinet G3 earthing switch switch-off	2
Medium voltage cabinet G3 energy storage coil not charged	3

Medium voltage cabinet G3 circuit breaker remote control	4
Medium voltage cabinet G1 load switch switch-on	5
Medium voltage cabinet G1 load switch switch-off	6
Medium voltage cabinet G1 earthing switch switch-on	7
Medium voltage cabinet G1 earthing switch switch-off	8
Medium voltage cabinet G2 load switch switch-on	9
Medium voltage cabinet G2 load switch switch-off	10
Medium voltage cabinet G2 earthing switch switch-on	11
Medium voltage cabinet G2 earthing switch switch-off	12
Low voltage room LA operation door open	13
Low voltage room LB operation door open	14
Medium voltage room door open	15

Appendix 19 PSS teleindication -- 5

Teleindication Description	Bit
Medium voltage cabinet G1 cable L1 phase over-temperature	0
Medium voltage cabinet G1 cable L2 phase over-temperature	1
Medium voltage cabinet G1 cable L3 phase over-temperature	2
Medium voltage cabinet G2 cable L1 phase over-temperature	3
Medium voltage cabinet G2 cable L2 phase over-temperature	4
Medium voltage cabinet G2 cable L3 phase over-temperature	5

Medium voltage cabinet G3 cable L1 phase over-temperature	6
Medium voltage cabinet G3 cable L2 phase over-temperature	7
Medium voltage cabinet G3 cable L3 phase over-temperature	8
Medium voltage protection general protection	9
Medium voltage protection general alarm	10

Appendix 20 PID alarm code -- 1

Alarm Description	Bit
Software version mismatch	0
Software and hardware version mismatch	1
Startup failure	2
Insulation resistance alarm	3
Insulation resistance pre-alarm	4
Over-temperature	5
Power module abnormal	6
Fan fault	7
Reserved	8
Inverter bus over-voltage protection	9
Output over-voltage protection	10
Inverter output over-voltage protection	11
Inverter output over-current protection	12

Output over-current protection	13
Output failure	14
Rs485 communication abnormal	15

Appendix 21 PID alarm code -- 2

Alarm Description	Bit
Grid power loss	0
Grid over-voltage	1
Grid under-voltage	2
Grid over-frequency	3
Grid under-frequency	4